4 Implement SQL Functions: Conversion Functions, and Date Functions.

Input:

```
CREATE TABLE employees (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(30),
    department VARCHAR(20),
    city VARCHAR(20),
    salary INT,
    commission INT,
    join_date DATE
);

INSERT INTO employees VALUES
(1, 'Anil',   'IT',      'Bengaluru', 50000, 5000, '2020-01-15'),
(2, 'Bhavna', 'HR',      'Mysuru',    40000, NULL, '2019-06-10'),
(3, 'Charan', 'IT',      'Mumbai',    60000, 8000, '2021-03-25'),
(4, 'Divya',  'Finance', 'Bengaluru', 45000, NULL, '2018-11-05'),
(5, 'Esha',   'Finance', 'Chennai',   55000, 3000, '2022-08-20');

CREATE TABLE contractors (
    con_id   INT PRIMARY KEY,
    con_name VARCHAR(30),
    city     VARCHAR(20)
);

INSERT INTO contractors VALUES
(1, 'Farhan', 'Mumbai'),
(2, 'Gita',   'Bengaluru'),
(3, 'Harsh',  'Delhi');

SELECT CAST('123' AS UNSIGNED) AS string_to_number;
SELECT CAST(salary AS CHAR) AS salary_as_text FROM employees;
SELECT CAST('2024-05-20' AS DATE) AS string_to_date;

SELECT CONVERT('456', UNSIGNED) AS string_to_number;
SELECT CONVERT(join_date, CHAR) AS date_as_text FROM employees;

SELECT STR_TO_DATE('20-05-2024', '%d-%m-%Y') AS converted_date;
```

```sql
SELECT emp_name, DATE_FORMAT(join_date, '%d-%m-%Y') AS formatted_date
FROM employees;

SELECT NOW()      AS current_datetime,
       CURDATE() AS current_date_only,
       CURTIME() AS current_time_only;

SELECT emp_name,
       join_date,
       YEAR(join_date)      AS join_year,
       MONTH(join_date)     AS join_month,
       DAY(join_date)       AS join_day,
       DAYNAME(join_date)   AS day_name,
       MONTHNAME(join_date) AS month_name
FROM employees;

SELECT emp_name,
       join_date,
       DATE_ADD(join_date, INTERVAL 10 DAY)   AS plus_10_days,
       DATE_ADD(join_date, INTERVAL 3 MONTH)  AS plus_3_months,
       DATE_SUB(join_date, INTERVAL 1 YEAR)   AS minus_1_year
FROM employees;

SELECT emp_name,
       DATEDIFF(CURDATE(), join_date)                    AS days_worked,
       TIMESTAMPDIFF(YEAR, join_date, CURDATE())         AS years_worked
FROM employees;

SELECT emp_name, join_date, LAST_DAY(join_date) AS month_end
FROM employees;
```

OUTPUT:

Output:

```
+------------------+
| string_to_number |
+------------------+
|              100 |
+------------------+
+-----------------+
| salary_as_text |
+-----------------+
| 50000          |
| 40000          |
| 60000          |
| 45000          |
| 55000          |
+-----------------+
+----------------+
| string_to_date |
+----------------+
| 2024-05-20     |
+----------------+
+------------------+
| string_to_number |
+------------------+
|              456 |
+------------------+
+--------------+
| date_as_text |
```

Output:

```
+--------------+
| date_as_text |
+--------------+
| 2020-01-15   |
| 2019-06-10   |
| 2021-03-25   |
| 2018-11-05   |
| 2022-08-20   |
+--------------+
+----------------+
| converted_date |
+----------------+
| 2024-05-20     |
+----------------+
+----------+----------------+
| emp_name | formatted_date |
+----------+----------------+
| Anil     | 15-01-2020     |
| Bhavna   | 10-06-2019     |
| Charan   | 25-03-2021     |
| Divya    | 05-11-2018     |
| Esha     | 20-08-2022     |
+----------+----------------+
+---------------------+--------------------+--------------------+
| current_datetime    | current_date_only | current_time_only |
+---------------------+--------------------+--------------------+
| 2026-10-02 16:02:07 | 2026-10-02         | 16:02:07           |
```

```
+----------+------------+-----------+------------+----------+-----------+------------+
| emp_name | join_date  | join_year | join_month | join_day | day_name  | month_name |
+----------+------------+-----------+------------+----------+-----------+------------+
| Anil     | 2020-01-15 |      2020 |          1 |       15 | Wednesday | January    |
| Bhavna   | 2019-06-10 |      2019 |          6 |       10 | Monday    | June       |
| Charan   | 2021-03-25 |      2021 |          3 |       25 | Thursday  | March      |
| Divya    | 2018-11-05 |      2018 |         11 |        5 | Monday    | November   |
| Esha     | 2022-08-20 |      2022 |          8 |       20 | Saturday  | August     |
+----------+------------+-----------+------------+----------+-----------+------------+
+----------+------------+--------------+---------------+--------------+
| emp_name | join_date  | plus_10_days | plus_3_months | minus_1_year |
+----------+------------+--------------+---------------+--------------+
| Anil     | 2020-01-15 | 2020-01-25   | 2020-04-15    | 2019-01-15   |
| Bhavna   | 2019-06-10 | 2019-06-20   | 2019-09-10    | 2018-06-10   |
| Charan   | 2021-03-25 | 2021-04-04   | 2021-06-25    | 2020-03-25   |
| Divya    | 2018-11-05 | 2018-11-15   | 2019-02-05    | 2017-11-05   |
| Esha     | 2022-08-20 | 2022-08-30   | 2022-11-20    | 2021-08-20   |
+----------+------------+--------------+---------------+--------------+
+----------+-------------+--------------+
| emp_name | days_worked | years_worked |
+----------+-------------+--------------+
| Anil     |        2452 |            6 |
| Bhavna   |        2671 |            7 |
| Charan   |        2017 |            5 |
| Divya    |        2888 |            7 |
| Esha     |        1504 |            4 |
+----------+-------------+--------------+
```

```
+----------+-------------+--------------+
| emp_name | days_worked | years_worked |
+----------+-------------+--------------+
| Anil     |        2452 |            6 |
| Bhavna   |        2671 |            7 |
| Charan   |        2017 |            5 |
| Divya    |        2888 |            7 |
| Esha     |        1504 |            4 |
+----------+-------------+--------------+
+----------+------------+------------+
| emp_name | join_date  | month_end  |
+----------+------------+------------+
| Anil     | 2020-01-15 | 2020-01-31 |
| Bhavna   | 2019-06-10 | 2019-06-30 |
| Charan   | 2021-03-25 | 2021-03-31 |
| Divya    | 2018-11-05 | 2018-11-30 |
| Esha     | 2022-08-20 | 2022-08-31 |
+----------+------------+------------+
```

5 Implement SQL Operators: Demonstrate the use of di”erent types of operators in SQL with examples, including Arithmetic Operators, Logical Operators, Comparison Operators

INPUT:

```
CREATE TABLE employees (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(30),
    department VARCHAR(20),
    city VARCHAR(20),
    salary INT,
    commission INT,
    join_date DATE
);

INSERT INTO employees VALUES
(1, 'Anil',   'IT',      'Bengaluru', 50000, 5000, '2020-01-15'),
(2, 'Bhavna', 'HR',      'Mysuru',    40000, NULL, '2019-06-10'),
(3, 'Charan', 'IT',      'Mumbai',    60000, 8000, '2021-03-25'),
(4, 'Divya',  'Finance', 'Bengaluru', 45000, NULL, '2018-11-05'),
(5, 'Esha',   'Finance', 'Chennai',   55000, 3000, '2022-08-20');

CREATE TABLE contractors (
    con_id   INT PRIMARY KEY,
    con_name VARCHAR(30),
    city     VARCHAR(20)
);

INSERT INTO contractors VALUES
(1, 'Farhan', 'Mumbai'),
(2, 'Gita',   'Bengaluru'),
(3, 'Harsh',  'Delhi');

SELECT 10 + 5  AS addition,
       10 - 5  AS subtraction,
       10 * 5  AS multiplication,
       10 / 4  AS division,
       10 DIV 4 AS integer_division,
       10 % 3  AS modulus;

SELECT emp_name,
```

```
       salary,
       salary + 5000        AS salary_with_bonus,
       salary * 12          AS yearly_salary,
       salary - (salary*0.1) AS salary_after_10pct_tax
FROM employees;
SELECT * FROM employees WHERE salary = 50000;
SELECT * FROM employees WHERE salary <> 50000;
SELECT * FROM employees WHERE salary > 45000;
SELECT * FROM employees WHERE salary < 50000;
SELECT * FROM employees WHERE salary >= 55000;
SELECT * FROM employees WHERE salary <= 45000;

SELECT * FROM employees
WHERE department = 'IT' AND salary > 50000;

SELECT * FROM employees
WHERE department = 'HR' OR department = 'Finance';

SELECT * FROM employees
WHERE NOT department = 'IT';

SELECT * FROM employees
WHERE (department = 'IT' OR department = 'Finance')
  AND salary >= 50000;
```

OUTPUT:

Output:

addition	subtraction	multiplication	division	integer_division	modulus
15	5	50	2.5000	2	1

emp_name	salary	salary_with_bonus	yearly_salary	salary_after_10pct_tax
Anil	50000	55000	600000	45000.0
Bhavna	40000	45000	480000	36000.0
Charan	60000	65000	720000	54000.0
Divya	45000	50000	540000	40500.0
Esha	55000	60000	660000	49500.0

emp_id	emp_name	department	city	salary	commission	join_date
1	Anil	IT	Bengaluru	50000	5000	2020-01-15

emp_id	emp_name	department	city	salary	commission	join_date
2	Bhavna	HR	Mysuru	40000	NULL	2019-06-10
3	Charan	IT	Mumbai	60000	8000	2021-03-25
4	Divya	Finance	Bengaluru	45000	NULL	2018-11-05
5	Esha	Finance	Chennai	55000	3000	2022-08-20

Output:

```
|      5 | Esha     | Finance    | Chennai   |  55000 |       3000 | 2022-08-20 |
+--------+----------+------------+-----------+--------+------------+------------+
+--------+----------+------------+-----------+--------+------------+------------+
| emp_id | emp_name | department | city      | salary | commission | join_date  |
+--------+----------+------------+-----------+--------+------------+------------+
|      1 | Anil     | IT         | Bengaluru |  50000 |       5000 | 2020-01-15 |
|      3 | Charan   | IT         | Mumbai    |  60000 |       8000 | 2021-03-25 |
|      5 | Esha     | Finance    | Chennai   |  55000 |       3000 | 2022-08-20 |
+--------+----------+------------+-----------+--------+------------+------------+
+--------+----------+------------+-----------+--------+------------+------------+
| emp_id | emp_name | department | city      | salary | commission | join_date  |
+--------+----------+------------+-----------+--------+------------+------------+
|      2 | Bhavna   | HR         | Mysuru    |  40000 |       NULL | 2019-06-10 |
|      4 | Divya    | Finance    | Bengaluru |  45000 |       NULL | 2018-11-05 |
+--------+----------+------------+-----------+--------+------------+------------+
+--------+----------+------------+---------+--------+------------+------------+
| emp_id | emp_name | department | city    | salary | commission | join_date  |
+--------+----------+------------+---------+--------+------------+------------+
|      3 | Charan   | IT         | Mumbai  |  60000 |       8000 | 2021-03-25 |
|      5 | Esha     | Finance    | Chennai |  55000 |       3000 | 2022-08-20 |
+--------+----------+------------+---------+--------+------------+------------+
+--------+----------+------------+-----------+--------+------------+------------+
| emp_id | emp_name | department | city      | salary | commission | join_date  |
+--------+----------+------------+-----------+--------+------------+------------+
|      2 | Bhavna   | HR         | Mysuru    |  40000 |       NULL | 2019-06-10 |
|      4 | Divya    | Finance    | Bengaluru |  45000 |       NULL | 2018-11-05 |
```

```
+--------+----------+------------+--------+--------+------------+------------+
| emp_id | emp_name | department | city   | salary | commission | join_date  |
+--------+----------+------------+--------+--------+------------+------------+
|      3 | Charan   | IT         | Mumbai |  60000 |       8000 | 2021-03-25 |
+--------+----------+------------+--------+--------+------------+------------+
```

```
+--------+----------+------------+-----------+--------+------------+------------+
| emp_id | emp_name | department | city      | salary | commission | join_date  |
+--------+----------+------------+-----------+--------+------------+------------+
|      2 | Bhavna   | HR         | Mysuru    |  40000 |       NULL | 2019-06-10 |
|      4 | Divya    | Finance    | Bengaluru |  45000 |       NULL | 2018-11-05 |
|      5 | Esha     | Finance    | Chennai   |  55000 |       3000 | 2022-08-20 |
+--------+----------+------------+-----------+--------+------------+------------+
```

```
+--------+----------+------------+-----------+--------+------------+------------+
| emp_id | emp_name | department | city      | salary | commission | join_date  |
+--------+----------+------------+-----------+--------+------------+------------+
|      2 | Bhavna   | HR         | Mysuru    |  40000 |       NULL | 2019-06-10 |
|      4 | Divya    | Finance    | Bengaluru |  45000 |       NULL | 2018-11-05 |
|      5 | Esha     | Finance    | Chennai   |  55000 |       3000 | 2022-08-20 |
+--------+----------+------------+-----------+--------+------------+------------+
```

```
+--------+----------+------------+-----------+--------+------------+------------+
| emp_id | emp_name | department | city      | salary | commission | join_date  |
+--------+----------+------------+-----------+--------+------------+------------+
|      1 | Anil     | IT         | Bengaluru |  50000 |       5000 | 2020-01-15 |
|      3 | Charan   | IT         | Mumbai    |  60000 |       8000 | 2021-03-25 |
|      5 | Esha     | Finance    | Chennai   |  55000 |       3000 | 2022-08-20 |
+--------+----------+------------+-----------+--------+------------+------------+
```

6 Implement SQL Operators: Demonstrate the use of di”erent types of operators in SQL with examples Special Operators, and Set Operations(UNION)

INPUT:

```
CREATE TABLE employees (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(30),
    department VARCHAR(20),
    city VARCHAR(20),
    salary INT,
    commission INT,
    join_date DATE
);

INSERT INTO employees VALUES
(1, 'Anil',   'IT',      'Bengaluru', 50000, 5000, '2020-01-15'),
(2, 'Bhavna', 'HR',      'Mysuru',    40000, NULL, '2019-06-10'),
(3, 'Charan', 'IT',      'Mumbai',    60000, 8000, '2021-03-25'),
```

```
(4, 'Divya',  'Finance', 'Bengaluru', 45000, NULL, '2018-11-05'),
(5, 'Esha',   'Finance', 'Chennai',   55000, 3000, '2022-08-20');

CREATE TABLE contractors (
    con_id   INT PRIMARY KEY,
    con_name VARCHAR(30),
    city     VARCHAR(20)
);

INSERT INTO contractors VALUES
(1, 'Farhan', 'Mumbai'),
(2, 'Gita',   'Bengaluru'),
(3, 'Harsh',  'Delhi');

SELECT * FROM employees
WHERE salary BETWEEN 45000 AND 55000;

SELECT * FROM employees
WHERE salary NOT BETWEEN 45000 AND 55000;

SELECT * FROM employees
WHERE city IN ('Bengaluru', 'Mumbai');

SELECT * FROM employees
WHERE city NOT IN ('Bengaluru', 'Mumbai');

SELECT * FROM employees WHERE emp_name LIKE 'A%';
SELECT * FROM employees WHERE emp_name LIKE '%a';
SELECT * FROM employees WHERE emp_name LIKE '_h%';

SELECT * FROM employees WHERE commission IS NULL;
SELECT * FROM employees WHERE commission IS NOT NULL;

SELECT emp_name FROM employees e
WHERE EXISTS (SELECT 1 FROM contractors c WHERE c.city = e.city);

SELECT * FROM employees
WHERE salary > ALL (SELECT salary FROM employees WHERE department = 'HR');

SELECT city FROM employees
```

```
UNION
SELECT city FROM contractors;

SELECT city FROM employees
UNION ALL
SELECT city FROM contractors;

SELECT emp_name AS name, city, 'Employee' AS type FROM employees
UNION
SELECT con_name, city, 'Contractor' FROM contractors;
```

OUTPUT:

Output:

emp_id	emp_name	department	city	salary	commission	join_date
1	Anil	IT	Bengaluru	50000	5000	2020-01-15
4	Divya	Finance	Bengaluru	45000	NULL	2018-11-05
5	Esha	Finance	Chennai	55000	3000	2022-08-20

emp_id	emp_name	department	city	salary	commission	join_date
2	Bhavna	HR	Mysuru	40000	NULL	2019-06-10
3	Charan	IT	Mumbai	60000	8000	2021-03-25

emp_id	emp_name	department	city	salary	commission	join_date
1	Anil	IT	Bengaluru	50000	5000	2020-01-15
3	Charan	IT	Mumbai	60000	8000	2021-03-25
4	Divya	Finance	Bengaluru	45000	NULL	2018-11-05

emp_id	emp_name	department	city	salary	commission	join_date
2	Bhavna	HR	Mysuru	40000	NULL	2019-06-10
5	Esha	Finance	Chennai	55000	3000	2022-08-20

```
+--------+----------+------------+---------+--------+------------+------------+
+--------+----------+------------+-----------+--------+------------+------------+
| emp_id | emp_name | department | city      | salary | commission | join_date  |
+--------+----------+------------+-----------+--------+------------+------------+
|      1 | Anil     | IT         | Bengaluru |  50000 |       5000 | 2020-01-15 |
+--------+----------+------------+-----------+--------+------------+------------+
+--------+----------+------------+-----------+--------+------------+------------+
| emp_id | emp_name | department | city      | salary | commission | join_date  |
+--------+----------+------------+-----------+--------+------------+------------+
|      2 | Bhavna   | HR         | Mysuru    |  40000 |       NULL | 2019-06-10 |
|      4 | Divya    | Finance    | Bengaluru |  45000 |       NULL | 2018-11-05 |
|      5 | Esha     | Finance    | Chennai   |  55000 |       3000 | 2022-08-20 |
+--------+----------+------------+-----------+--------+------------+------------+
+--------+----------+------------+--------+--------+------------+------------+
| emp_id | emp_name | department | city   | salary | commission | join_date  |
+--------+----------+------------+--------+--------+------------+------------+
|      2 | Bhavna   | HR         | Mysuru |  40000 |       NULL | 2019-06-10 |
|      3 | Charan   | IT         | Mumbai |  60000 |       8000 | 2021-03-25 |
+--------+----------+------------+--------+--------+------------+------------+
+--------+----------+------------+-----------+--------+------------+------------+
| emp_id | emp_name | department | city      | salary | commission | join_date  |
+--------+----------+------------+-----------+--------+------------+------------+
|      2 | Bhavna   | HR         | Mysuru    |  40000 |       NULL | 2019-06-10 |
|      4 | Divya    | Finance    | Bengaluru |  45000 |       NULL | 2018-11-05 |
+--------+----------+------------+-----------+--------+------------+------------+
+--------+----------+------------+-----------+--------+------------+------------+
| emp_id | emp_name | department | city      | salary | commission | join_date  |
```

emp_id	emp_name	department	city	salary	commission	join_date
1	Anil	IT	Bengaluru	50000	5000	2020-01-15
3	Charan	IT	Mumbai	60000	8000	2021-03-25
5	Esha	Finance	Chennai	55000	3000	2022-08-20

emp_name
Anil
Charan
Divya

emp_id	emp_name	department	city	salary	commission	join_date
1	Anil	IT	Bengaluru	50000	5000	2020-01-15
3	Charan	IT	Mumbai	60000	8000	2021-03-25
4	Divya	Finance	Bengaluru	45000	NULL	2018-11-05
5	Esha	Finance	Chennai	55000	3000	2022-08-20

city
Bengaluru

Output:

```
+-----------+
| city      |
+-----------+
| Bengaluru |
| Mysuru    |
| Mumbai    |
| Chennai   |
| Delhi     |
+-----------+
+-----------+
| city      |
+-----------+
| Bengaluru |
| Mysuru    |
| Mumbai    |
| Bengaluru |
| Chennai   |
| Mumbai    |
| Bengaluru |
| Delhi     |
+-----------+
+--------+-----------+------------+
| name   | city      | type       |
+--------+-----------+------------+
| Anil   | Bengaluru | Employee   |
| Bhavna | Mysuru    | Employee   |
| Charan | Mumbai    | Employee   |
```

name	city	type
Anil	Bengaluru	Employee
Bhavna	Mysuru	Employee
Charan	Mumbai	Employee
Divya	Bengaluru	Employee
Esha	Chennai	Employee
Farhan	Mumbai	Contractor
Gita	Bengaluru	Contractor
Harsh	Delhi	Contractor